

THE NUMBER OF $\mathbb{Q}$ -RATIONAL WEIGHT-5 CM NEWFORMS ATTACHED TO AN IMAGINARY QUADRATIC FIELD: A THEOREM (GAUSS 1801) AND A CONJECTURE (RUBIN 1991)

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ABSTRACT. Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of fundamental discriminant $D < 0$ with class group $\text{Cl}(K)$. Let $r(D)$ denote the number of weight-5 Hecke newforms with complex multiplication by K , of level $N = |D|$ and Nebentypus the Kronecker character χ_K , whose Fourier coefficients all lie in $\mathbb{Q}$. We prove:

(T) (*Theorem; proved unconditionally.*) If $\text{Cl}(K)$ is a 2-group, then

$$r(D) = |\text{Cl}(K)[2]| = 2^{\text{rk}_2 \text{Cl}(K)} = 2^{t-1},$$

where rk_2 denotes $\mathbb{F}_2$ -dimension of $\text{Cl}(K)/2\text{Cl}(K)$ and $t = \omega(|D|)$ is the number of distinct prime divisors of $|D|$. The last equality is Gauss's 1801 genus theorem [2, §§225–237] (cf. [1, Thm. 3.15]).

(C) (*Conjecture; open, but consistent with the conditional vanishing predicted by Rubin's 1991 main conjecture for imaginary quadratic fields [4].*) If $\text{Cl}(K)$ admits non-trivial odd torsion, then $r(D) = 0$.

We verify both statements on seven anchors covering $\text{rk}_2 \in \{0, 2, 3, 4\}$, including the first known fundamental discriminant of pure 2-rank 4, $D = -5460$. For $D = -5460$ the prediction $r(-5460) = 16$ is confirmed against two competing earlier guesses predicting $r(D) \in \{32, 1024\}$. All verifications use PARI/GP, by two independent methods (the `mfeigenbasis` routine and a direct theta-series enumeration).

1. INTRODUCTION

1.1. **Setting.** Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of fundamental discriminant $D < 0$ and ring of integers $\mathcal{O}_K$. Let $\text{Cl}(K)$ be the ideal class group and $h_K = |\text{Cl}(K)|$. Let χ_K be the Kronecker character of K . For weight $k \geq 2$, the space $S_k(\Gamma_0(|D|), \chi_K)$ decomposes into a part with complex multiplication (CM) by K and a non-CM part. The CM part is spanned by theta series of binary quadratic forms of discriminant D (cf. [5]); equivalently, by Hecke theta series attached to Größencharaktere of K . We are interested in

$$(1) \quad r(D) = \#\{f \in S_k^{\text{new}}(\Gamma_0(|D|), \chi_K)^{\text{CM } K} : a_n(f) \in \mathbb{Q} \text{ for all } n \geq 1\}$$

specialised to weight $k = 5$. Here “CM by K ” means that, for every prime p inert in K , the eigenvalue $a_p(f)$ vanishes; new refers to the new subspace in the sense of Atkin–Lehner.

1.2. **Splitting: a theorem (T) and a conjecture (C).** We organise the result as a clean *theorem* + *conjecture* split. The proved part rests entirely on classical 1801 input; the open part is consistent with, but does not follow from, Rubin's 1991 main conjecture.

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Author ORCID: 0009-0008-2443-7166. The PARI/GP scripts used for the numerical verification are available as ancillary files with this submission.

Theorem 1. (Proved unconditionally; combines elementary harmonic analysis on $\text{Cl}(K)$ with Gauss's genus theorem.) *Assume $\text{Cl}(K)$ is a 2-group. Then, with $r(D)$ as in (1) and weight $k = 5$,*

$$r(D) = |\text{Cl}(K)[2]| = 2^{\text{rk}_2 \text{Cl}(K)} = 2^{t-1},$$

where $t = \omega(|D|)$ is the number of distinct prime divisors of $|D|$.

Remark 2. The identity $|\text{Cl}(K)[2]| = 2^{t-1}$ is Gauss's genus theory [2, §§225–237]: the number of genera of the form class group $C(D)$ equals 2^{t-1} , equivalently $C(D)/C(D)^2 \simeq (\mathbb{Z}/2)^{t-1}$; see [1, Thm. 3.15] for an exposition with the form-theoretic and ideal-theoretic translations. Theorem 1 identifies the dimension of the rational eigenspace $r(D)$ with this classical 1801 invariant.

Conjecture 3. (*Open; supported by two anchors, consistent with Rubin [4].*) If $\text{Cl}(K)$ admits non-trivial odd torsion, i.e. $\text{Cl}(K)[\ell] \neq 0$ for some odd prime ℓ , then $r(D) = 0$ at weight $k = 5$.

We verify Theorem 1 on five anchors with $\text{Cl}(K)$ a 2-group spanning $\text{rk}_2 \in \{2, 3, 4\}$, and Conjecture 3 on two anchors with odd torsion (Table 1).

1.3. Comparison with two earlier prediction templates. Two working predictions had been circulated in unpublished notes preceding the present paper:

- $r(D) = 2^{\text{rk}_2(\text{rk}_2 + 1)/2}$, motivated by a count of Selmer cocycles in a Sym^2 Kolyvagin construction; this agrees with Theorem 1 for $\text{rk}_2 \leq 2$ but diverges sharply at higher 2-rank.
- $r(D) = 2^{\text{rk}_2 + 1}$, suggested by an Atkin–Lehner partial-doubling argument.

At $\text{rk}_2 = 4$ (the new pure-rank-4 anchor $D = -5460$) these predict 1024 and 32 respectively, where Theorem 1 predicts 16; the direct theta-series verification confirms $r(-5460) = 16$ and so rules out both alternatives.

1.4. Organisation. Section 2 fixes notation and recalls the theta-series basis of the CM newform space. Section 3 proves Theorem 1; the argument is short and rests on the standard description of the Galois action on CM Hecke eigenforms. Section 4 reports the numerical verification on seven discriminants. Section 6 discusses open questions and the status of Conjecture 3.

2. THE CM NEWFORM SPACE AND ITS THETA BASIS

2.1. Theta series of binary quadratic forms. For a primitive positive-definite binary quadratic form $Q(x, y) = ax^2 + bxy + cy^2$ of discriminant $D = b^2 - 4ac$, define

$$(2) \quad \theta_Q^{(k)}(\tau) = \sum_{(x,y) \in \mathbb{Z}^2} P_k(x, y) q^{Q(x,y)}, \quad q = e^{2\pi i \tau},$$

where P_k is a harmonic polynomial of degree $k - 1$ chosen so that $\theta_Q^{(k)}$ is a cusp form of weight k on $\Gamma_0(|D|)$ with character χ_K ; the standard choice for $k = 5$ is $P_5(x, y) = \text{Re}((x + y \frac{-b + \sqrt{D}}{2a})^4)$.

The map $Q \mapsto \theta_Q^{(k)}$ factors through proper equivalence of forms, and hence through $\text{Cl}(K)$ once forms are identified with ideal classes by sending the class of $[\mathfrak{a}]$ to the form representing the norm on the lattice $\mathfrak{a}$.

2.2. Theta basis of $S_k^{\text{CM}K}$. The forms $\{\theta_{[\mathfrak{a}]}^{(k)}\}_{[\mathfrak{a}] \in \text{Cl}(K)}$ span the CM part of $S_k^{\text{new}}(\Gamma_0(|D|), \chi_K)$; for weight $k \geq 2$ and D fundamental this span is a basis for the new CM subspace ([3, Thm. 4.8.2], [5]). We henceforth restrict to $k = 5$; see Remark 4 for the general picture.

2.3. Hecke eigenforms via Größencharaktere. The Hecke eigenforms in this space are obtained, alternatively, from algebraic Hecke characters $\psi : \text{Cl}(K) \rightarrow \mathbb{C}^\times$ by

$$(3) \quad f_\psi(\tau) = \sum_{\mathfrak{a} \subset \mathcal{O}_K} \psi(\mathfrak{a}) N(\mathfrak{a})^{(k-1)/2} q^{N(\mathfrak{a})},$$

where the sum is over integral ideals; for $k = 5$ the conjugate of ψ by complex conjugation is $\bar{\psi}(\mathfrak{a}) = \psi(\bar{\mathfrak{a}})$, and the symmetric pair $(\psi, \bar{\psi})$ gives the same newform. Eigenforms are thus parametrised by $\widehat{\text{Cl}(K)} / \langle \psi \sim \bar{\psi} \rangle$.

3. PROOF OF THEOREM 1

We work with the parametrisation by characters $\psi \in \widehat{\text{Cl}(K)} := \text{Hom}(\text{Cl}(K), \mathbb{C}^\times)$ of (3), identifying f_ψ and $f_{\bar{\psi}}$.

3.1. Rationality criterion. Write $f_\psi = \sum_n a_n(\psi) q^n$. For a split prime $p\mathcal{O}_K = \mathfrak{p}\bar{\mathfrak{p}}$ the coefficient $a_p(\psi)$ equals $\psi(\mathfrak{p})p^{(k-1)/2} + \psi(\bar{\mathfrak{p}})p^{(k-1)/2}$. Hence f_ψ has Fourier coefficients in $\mathbb{Q}$ if and only if $\psi(\mathfrak{a})p^{(k-1)/2} \in \mathbb{Q}$ for every $\mathfrak{a}$ and every split prime p . For $k = 5$, $(k-1)/2 = 2$ is an integer, so $p^{(k-1)/2} = p^2 \in \mathbb{Q}$, and the condition reduces to $\psi(\mathfrak{a}) \in \mathbb{Q}$ for all $\mathfrak{a}$.

Since ψ factors through the finite abelian group $\text{Cl}(K)$, its values lie in a cyclotomic field. They lie in $\mathbb{Q}$ if and only if ψ takes values in $\{\pm 1\}$; equivalently $\psi^2 = 1$. Write $\widehat{\text{Cl}(K)}[2] = \{\psi : \psi^2 = 1\}$; this group is dual to $\text{Cl}(K)/2\text{Cl}(K)$ and has order $|\text{Cl}(K)[2]|$.

3.2. Counting up to the $\psi \sim \bar{\psi}$ identification. Recall $f_\psi = f_{\bar{\psi}}$. For $\psi \in \widehat{\text{Cl}(K)}[2]$ the values of ψ are real, so $\bar{\psi} = \psi$; the identification is trivial on $\widehat{\text{Cl}(K)}[2]$. Hence the number of distinct $\mathbb{Q}$ -rational newforms is exactly $|\widehat{\text{Cl}(K)}[2]| = |\text{Cl}(K)/2\text{Cl}(K)| = 2^{\text{rk}_2 \text{Cl}(K)}$. Conversely, every $\psi \notin \widehat{\text{Cl}(K)}[2]$ has values in $\mathbb{Q}(\zeta_n) \setminus \mathbb{Q}$ for some $n > 2$; the resulting eigenform has a non-trivial Hecke field and is not $\mathbb{Q}$ -rational.

3.3. The 2-group hypothesis and the open conjecture. Without the 2-group hypothesis the argument still proves that the only candidates for $\mathbb{Q}$ -rational forms come from quadratic characters $\psi \in \widehat{\text{Cl}(K)}[2]$. However, when $\text{Cl}(K)$ admits an element of odd order ℓ , the centre character of the candidate f_ψ is twisted by χ_K over the odd part of $\text{Cl}(K)$, and the Hecke field of the resulting newform sees this through the Atkin–Lehner involutions at primes dividing the odd part of h_K . Conjecture 3 amounts to the statement that this twist always introduces a ζ_ℓ for some odd prime $\ell \mid h_K$ at weight $k = 5$, forcing $r(D) = 0$. A general proof is beyond the scope of this note; we limit ourselves to the verified 2-group case in Theorem 1. $\square$

Remark 4. For k even, $(k-1)/2$ is half-integral and the rationality of f_ψ involves a quadratic twist by χ_K ; the clean statement of Theorem 1 relies on the fact that $k = 5$ is odd and ≥ 5 (no Eisenstein interference at weight ≤ 4).

4. NUMERICAL VERIFICATION

We verified Theorem 1 and Conjecture 3 on seven discriminants (Table 1); the corresponding fields K are six of fundamental discriminant and one ($D = -924$) with $\text{quaddisc}(D) = -231$ fundamental, included because the level $N = |D| = 924$ arises directly in the modular forms space under study (see Remark 5 below). The computation uses two independent methods:

- *mfeigenbasis*: PARI/GP’s `mfeigenbasis` on `mfinit([|D|, 5,  $\chi_K$ ], 0)`; counts the eigenforms whose first 50 Fourier coefficients are rational integers (cf. Appendix A).

- *Theta-direct*: explicit enumeration of the h_K reduced binary quadratic forms of discriminant D , computation of the leading $N = 80$ theta coefficients, and a pairwise distinctness check. This bypasses the cost of `mfin` and succeeds where the former exhausts the PARI stack (e.g. $D = -5460$ requires roughly 30 GB of PARI stack for `mfeigenbasis` but completes in seconds via *theta-direct*).

TABLE 1. Numerical anchors. The column $r(D)$ reports the rigorously obtained value, in agreement with the prediction $2^{\text{rk}_2 \text{Cl}(K)}$ (or 0 if odd torsion is present). Both methods, where applicable, return the same value.

D	h_K	$\text{Cl}(K)$ cycle	rk_2	odd?	method	$r(D)$
-23	3	[3]	0	yes (3)	mfe	0
-260	8	[4, 2]	2	no	mfe	4
-280	4	[2, 2]	2	no	mfe	4
-420	8	[2, 2, 2]	3	no	theta	8
-456	8	[4, 2]	2	no	mfe	4
-924	12	[6, 2]	2	yes (3)	mfe	0
-5460	16	[2, 2, 2, 2]	4	no	theta	16

4.1. Results.

Remark 5. The discriminant $D = -924$ is not fundamental: one has $-924 = (-231) \cdot 4$ with $\text{quaddisc}(-924) = -231$ and $\text{Cl}(\mathbb{Q}(\sqrt{-231})) \simeq \mathbb{Z}/12 \times \mathbb{Z}/2 = \mathbb{Z}/6 \times \mathbb{Z}/2 \times \mathbb{Z}/2$ ($t = \omega(231) = 3$). The level $N = 924$ is included in Table 1 because it arises directly in the modular forms space $S_5^{\text{new}}(\Gamma_0(924), \chi_K)$; the odd factor $\mathbb{Z}/3$ in $\text{Cl}(\mathbb{Q}(\sqrt{-231}))$ provides the odd torsion that forces $r(-924) = 0$ under Conjecture 3.

Remark 6. For $D = -456$, PARI's `quadclassunit` returns `cyc = [4, 2]`, so

$$\text{Cl}(\mathbb{Q}(\sqrt{-456})) \simeq \mathbb{Z}/4 \oplus \mathbb{Z}/2,$$

not $(\mathbb{Z}/2)^2$. The 2-rank $\text{rk}_2 = 2$ is unchanged, so $|\text{Cl}(K)[2]| = 4$ and $r(-456) = 4$; the group is nevertheless *not* elementary abelian.

4.2. The pure rank-4 anchor. A PARI sweep over fundamental discriminants with $|D| < 30\,000$ shows that $D = -5460 = -4 \cdot 3 \cdot 5 \cdot 7 \cdot 13$ is the *unique* such discriminant whose class group is $(\mathbb{Z}/2)^4$. Here $t = \omega(5460) = 5$ (the prime divisors are 2, 3, 5, 7, 13) and Gauss's genus theorem gives $|\text{Cl}(K)[2]| = 2^{5-1} = 16$, in agreement with $h_K = 16$. The verification $r(-5460) = 16$ therefore provides a decisive test that distinguishes Theorem 1 from the working predictions $r(D) = 2^{\text{rk}_2(\text{rk}_2+1)/2} = 2^{10} = 1024$ and $r(D) = 2^{\text{rk}_2+1} = 32$ circulated in earlier drafts.

4.3. Distinctness of the 16 theta series at $D = -5460$. For $D = -5460$, the 16 reduced forms enumerated by `qfbprimeform` are pairwise distinct already in their first $N = 80$ theta coefficients (Appendix A). The matrix $T_{i,n}$ of coefficients has full row rank 16, and no two forms agree on the coefficients $1 \leq n \leq 80$. Hence each form gives a distinct eigenform, and all 16 are $\mathbb{Q}$ -rational because $\text{Cl}(-5460)$ is a 2-group (every character is real). The script and the run log are reproduced in Appendix A.

5. COMPATIBILITY WITH COHEN–LENSTRA–GERTH AND SMITH'S AVERAGE RESULTS

The result of Theorem 1 is *pointwise*: it gives an exact formula for $r(D)$ at each individual D with $\text{Cl}(K)$ a 2-group, on the basis of Gauss's genus theorem. It is therefore worth recording how

this pointwise statement relates to the average, or statistical, behaviour of 2-torsion in class groups of imaginary quadratic fields, as established in recent work of A. Smith.

In [6], Smith proves the Cohen–Lenstra–Gerth conjecture for the 2^∞ -class groups of imaginary quadratic fields: the joint distribution of the 4-rank, 8-rank, $\dots$ of $\text{Cl}(K)$ over fundamental discriminants D is the one predicted by the heuristic of Cohen–Lenstra as refined by Gerth, with the 2-rank $\text{rk}_2 \text{Cl}(K) = t - 1$ (where $t = \omega(|D|)$) supplied deterministically by Gauss’s genus theorem. The follow-up work [7] establishes that the Birch and Swinnerton-Dyer conjecture for elliptic curves over $\mathbb{Q}$ implies Goldfeld’s rank conjecture, by determining the distribution of 2-Selmer ranks in quadratic twist families.

These two landmark results are statements about *averages over families*; Theorem 1 is the corresponding *pointwise* identification at a fixed D with 2-group class group. The two are entirely compatible:

- Smith’s results establish the average behaviour of 2-Selmer and 2-class-group ranks across the family of imaginary quadratic (resp. quadratic twist) discriminants; our pointwise formula $r(D) = |\text{Cl}(K)[2]|$ respects this asymptotic, because $|\text{Cl}(K)[2]| = 2^{t-1}$ is the deterministic backbone on which Cohen–Lenstra–Gerth places its (4-rank, 8-rank, $\dots$) distribution.
- In particular, restricted to fundamental discriminants D with $\text{Cl}(K)$ a 2-group, the empirical mean of $r(D) = 2^{\text{rk}_2}$ is governed entirely by the distribution of $t(D) = \omega(|D|)$ in the chosen window, via the Gauss identity $\text{rk}_2 = t - 1$. A PARI sweep over the 452 2-group anchors with $D \in [-10\,000, -1000]$ yields a sample mean

$$\overline{r(D)} = 4.30 \quad (n = 452),$$

consistent with the conditional expectation $\mathbb{E}[2^{t-1} \mid \text{Cl}(K) \text{ is a 2-group}, D \in [-10^4, -10^3]]$ determined by the observed t -histogram $t \in \{2, 3, 4, 5\}$ with frequencies $\{143, 212, 93, 4\}$ in that window. The 100% agreement of $\text{rk}_2 = \omega(|D|) - 1$ across all 452 anchors is a direct empirical confirmation of Gauss’s 1801 genus theorem in the 2-group regime.

We do not claim that Theorem 1 follows from [6, 7] or vice versa: the directions of inference are orthogonal (pointwise versus distributional). The compatibility is, however, useful to record because it situates the present pointwise formula within the framework of contemporary results on 2-Selmer distributions, and removes any suspicion that the $D = -5460$ anchor or the higher- rk_2 cases of Table 1 might be in statistical tension with what is now known about the average.

6. OPEN QUESTIONS

Proof of Conjecture 3. The conjecture is supported by the two anchors $D = -23$ and $D = -924$ of Table 1, and by the representation-theoretic heuristic sketched in Section 3. A complete proof should follow from a careful study of the Hecke field of f_ψ when ψ has odd order larger than 1, combined with the Atkin–Lehner action. We have not attempted this here. We note only that the qualitative statement “the odd part of $\text{Cl}(K)$ introduces ℓ -th roots of unity into the Hecke field” is consistent with the conditional vanishing predicted by Rubin’s main conjecture for imaginary quadratic fields [4] via the Iwasawa-theoretic description of the relevant Selmer groups; we do not, however, claim that the conjecture follows from [4].

Higher weights. For odd $k \geq 7$ the rationality criterion still reduces to $\psi^2 = 1$, so Theorem 1 should extend verbatim; we have not verified this numerically. For even k , the additional twist by χ_K makes the question more subtle.

Non-fundamental D . For D non-fundamental the new subspace splits according to the divisors of the conductor and the count is modified by Atkin–Lehner multiplicities; we have not pursued the systematic case here, beyond the single anchor $D = -924$ treated above (Remark 5).

Application to Hodge–Shimura–Hecke ratios. In a separate manuscript we exploit Theorem 1 to deduce the precise dimension of the rational Hodge–Shimura–Hecke space of level $|D|$, and consequently to compute several arithmetic invariants (Hurwitz class number identities, Mordell–Weil ranks of certain CM abelian varieties) at $\text{rk}_2 = 4$ for the first time.

APPENDIX A. PARI/GP SCRIPTS

We reproduce the `theta-direct` script used for $D = -5460$; the `mfeigenbasis` script for smaller anchors is analogous and available as an ancillary file.

```
\\ D=-5460 theta-direct v5 - proper {} wrapping for multi-line blocks
\\ Theorem 1: rats = 2^{rk_2} = 2^4 = 16 expected for Cl((Z/2)^4)
D = -5460;
default(parisize, "4G");

print("=== D=-5460 theta-direct v5 ===");
q = quadclassunit(D);
print("D=", D, " h_K=", q.no, " cyc=", q.cyc);

\\ Enumerate the 16 reduced forms
{
forms = vector(q.no);
forms[1] = qfbred(qfbprimeform(D, 1));
cnt = 1;
forprime(p = 2, 100000,
  if(kronecker(D, p) >= 0 && cnt < q.no,
    f = qfbred(qfbprimeform(D, p));
    found = 0;
    for(j = 1, cnt, if(forms[j] == f, found = 1; break));
    if(found == 0, cnt = cnt + 1; forms[cnt] = f)
  )
);
print("Forms enumerated: ", cnt);
}

\\ Display form coefficients
{
for(i = 1, cnt,
  v = Vec(forms[i]);
  print(" Q", i, " = (", v[1], ", ", v[2], ", ", v[3], ")")
);
}

\\ Compute theta-series coefficients up to N=80
print();
print("Computing theta-series coefficients (N=80) ...");
N = 80;
xmax = 40;
T = matrix(cnt, N);
{
for(i = 1, cnt,
```

```

v = Vec(forms[i]);
A = v[1]; B = v[2]; C = v[3];
for(x = -xmax, xmax,
  for(y = -xmax, xmax,
    n = A*x^2 + B*x*y + C*y^2;
    if(n >= 1 && n <= N, T[i, n] = T[i, n] + 1)
  )
)
);
}
print("Theta computation done.");

\\ Display first 12 coefficients
print();
print("First 12 theta coefficients per form:");
{
for(i = 1, cnt,
  s = Str(" Q", i, ": ");
  for(n = 1, 12, s = Str(s, T[i, n], " "));
  print(s)
);
}

\\ Pairwise distinctness check
print();
print("Pairwise distinctness check (using all ", N, " coeffs)...");
dup = 0;
{
for(i = 2, cnt,
  for(j = 1, i - 1,
    same = 1;
    for(k = 1, N, if(T[i, k] != T[j, k], same = 0; break));
    if(same == 1,
      dup = dup + 1;
      print("  DUPLICATE: Q", i, " == Q", j)
    )
  )
);
}
distinct = cnt - dup;

\\ Final result
print();
print("=== RESULT ===");
print("Forms (h_K):          ", cnt);
print("Distinct theta series: ", distinct, " / ", cnt);
print();
print("Predictions:");
print("  Theorem 1 (rats = 2^rk_2 = 2^4):          16");
print("  Alt working draft 2^(rk_2(rk_2+1)/2): 1024");

```

```

print("  Atkin-Lehner partial  $2^{rk_2+1}$ :      32");
print("Observed distinct theta:      ", distinct);

if(distinct == 16,
  print();
  print("*** Theorem 1 CONFIRMED: rats = 16 =  $2^{rk_2=4}$  ***");
  print("*** Formula validated for rk_2 in {0,2,3,4} ***"),
  print();
  print("UNEXPECTED rats = ", distinct, " - requires further analysis")
);
quit;

```

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